QP CODE: 24900203

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

FIRST SEMESTER MGU-UGP (HONOURS) REGULAR EXAMINATION NOVEMBER 2024

Discipline Specific Core Course - MG1DSCPHY100 - FOUNDATIONS OF PHYSICS

Time:- $1\frac{1}{2}$ Hour **Marks:-50**

Part A

Short Answer Type Questions
Answer any 7 questions
Each question carries 2 marks

1. Q. Convert

   (a) 10 nanometers (nm) to meters (m)

   (b) 10 meters (m) to micrometers (μm). **CO-1, LD-U**

   Ans. (a) $10nm = 10^{-8}m$ **1 mark**

   (b) $10m = 10^{7}\mu m$ **1 mark**

2. Q. Two vectors **A** and **B** are Parallel.What will be their vector product and scalar product? **CO-1, LD-A**

   Ans. Both these vectors are parallel, $\therefore \theta = 0$

   (a) $\boldsymbol{A} \times \boldsymbol{B} = AB\sin(\theta)\hat{n} = 0$ **1 mark**

   (b) $\boldsymbol{A} \bullet \boldsymbol{B} = AB\cos(\theta) = AB$ **1 mark**

3. Q. Differentiate between average acceleration and instantaneous acceleration. **CO-2, LD-U**

   Ans. (a) Average Acceleration $= \dfrac{v_2 - v_1}{t_2 - t_1} = \dfrac{\Delta v}{\Delta t}$ **OR** definition **1 mark**

   (b) Instantaneous Acceleration. $= \lim_{\Delta t \to 0} \dfrac{\Delta v}{\Delta t} = \dfrac{dv}{dt}$ **OR** definition **1 mark**

4. Q. What is the horizontal component of velocity in projectile motion? **CO-3, LD-U**

   Ans. $U_x = U\cos(\theta)$ **OR** The velocity along the x direction (horizontal component of velocity) is a constant in time.

   2 mark

5. *Q*. (a) How is acceleration related to velocity of a particle?

(b) Will the acceleration of a particle become zero when it moves with a constant speed along a curved path? **CO-3, LD-U**

Ans. (a) Acceleration is the rate of change of velocity $a = \frac{dv}{dt}$ **or** $v = u + at$ **1 mark**

(b) No, Even though the speed is constant, the direction of the velocity changes continuously along the curve. **1 mark**

6. *Q*. A car is initially moving in reverse. The driver applies brakes, slowing the car. What is the direction of car's acceleration, relative to the car? **CO-3, LD-An**

Ans. The car is moving backward (reverse), the acceleration along the forward direction(opposite to its motion) **2 mark**

7. *Q*. What is the power delivered when work of 1000 J is done in 2 seconds. **CO-4, LD-A**

Ans. $P = \frac{W}{t}$ **1 mark**

$P = \frac{1000\text{J}}{2\text{s}} = 500\,\text{J/s(W)}$ **1 mark**

8. *Q*. What is the difference between positive work and negative work? **CO-4, LD-U**

Ans. Positive work occurs when the force (component) and displacement are in the same direction, while negative work happens when they are opposite. **2 mark**

9. *Q*. Assume that the earth moves around the sun in a circular orbit. Does the sun do any work on the earth? **CO-4, LD-An**

Ans. No, the Sun does not do any work on the Earth.
Work is given by the equation:

$$W = F \cdot d \cdot \cos(\theta) = F \cdot d \cdot \cos(90^\circ) = 0$$

$\therefore$ Work done at each and every point is zero. **2 mark**

10. *Q*. Write the syntax of if else statements in Python.

CO-5, LD-K

Ans.
```
if condition:
    # code to execute if the condition is True
else:
    # code to execute if the condition is False
```

2 mark

OR

Any meaningful example can be considered as correct answer **2 mark**

Part B

Short Essay Type Questions
Answer any 4 questions
Each question carries 6 marks

11. *Q*. Sketch and show the resultant **R** of vector sum of **A** and **B** **CO-1, LD-U**

Ans. Triangular or parallelogram laws of vector addition. **6 mark**

12. *Q*. A cheetah is found at $20m$ to the east of a vehicle. At time $t = 0$, it begins to run due east towards its prey which is at $50m$ to the east of the vehicle.During the first $2.0s$ of the chase, the Cheetah's x-coordinate varies with the time according to the equation $x = 20m + (5m/s^2)t^2$. Find the Cheetah's instantaneous velocity at $t_1 = 1$ s by taking, $\Delta t = 0.1s$, then $0.01s$ and $0.001s$. **CO-2, LD-A**

Ans. The velocity is the rate of change of position with respect to time.
$v(t) = \lim_{\Delta t \to 0} \frac{x(t+\Delta t) - x(t)}{\Delta t}$ **1.5 Marks**

$x(1) = 20 + (5)(1)^2 =$ 25 m $\quad$ $x(1+0.1) = 20 + 5(1.1)^2 =$ 26.05 m
$x(1+0.01) = 20 + 5(1.01)^2 =$ 25.1005 m $\quad$ $x(1+0.001) = 20 + 5(1.001)^2 =$ 25.010005 m

1.5 marks

$$\Delta t = 0.1\,\text{s} \;\; v(1) \approx \frac{26.05 - 25}{0.1} = 10.50\,\text{m/s}$$
$$\Delta t = 0.01\,\text{s} \;\; v(1) \approx \frac{25.1005 - 25}{0.01} = 10.05\,\text{m/s}$$
$$\Delta t = 0.001\,\text{s} \;\; v(1) \approx \frac{25.010005 - 25}{0.001} = 10.005\,\text{m/s}$$ **3 marks**

OR

$v = \frac{dx}{dt} = (10m/s^2)t$ at $t = 1 \quad v_{t=1} = 10m/s$ **6 Marks**

13. *Q*. An iceboat is at rest on a frictionless horizontal surface. Due to the blowing wind, $4.0s$ after the iceboat is released, it is moving to the right at $6.0m/s$(about 22 km/h, or 13 mi/h). What constant horizontal force does the wind exert on the iceboat? **CO-3, LD-A**

Ans. Mass of the object is not given in this question, so let as take it as "M".

$F = ma$ **2 marks**

$v = u + at$ **2 marks**

Where: Final velocity $v = 6.0$ m/s. Initial velocity $u = 0$ m/s. Time interval $t = 4.0$ s.
The average acceleration $a_{avg} = \frac{v-u}{t} = \frac{6.0\,\text{m/s} - 0}{4.0\,\text{s}} = \frac{6.0}{4.0} = 1.5\,\text{m/s}^2$ **2 marks**

$$\therefore F_{avg} = M \times 1.5\ \text{m/s}^2$$

14. *Q*. Discuss the transfer of energy in trampoline jumping. **CO-4, LD-An**

Ans. While a person jumps, potential energy converts to kinetic energy. Upon landing, kinetic energy transfers to the trampoline as elastic potential energy, which then propels the person back up. Some energy is lost as heat and sound. **6 marks**

OR

Any explanation relating Kinetic energy and Gravitational potential energy. **6 marks**

15. *Q*. What is the work done by a constant force? **CO-4, LD-U**

Ans. $W = \mathbf{F} \cdot \mathbf{d} = Fd\cos\theta$ **3 marks**

Where: $\mathbf{F}$ is the constant force. d is the displacement of the object. θ is the angle between the force and displacement. **3 marks**

16. *Q*. Write a Python program to determine if a given number is positive, negative, or zero. **CO-5, LD-A**

Ans. Python program with correct syntax **6 marks**

Part C

Essay Type Questions
Answer any 1 question
Each question carries 12 marks

17. *Q*. Explain the operators in Python and their rules of precedence. **CO-5, LD-U**

Ans. Explanation of basic mathematical operations with descriptions or example **10 marks**
Rules of precedence **2 marks**

18. *Q*. Give a detailed account on errors in python programming. **CO-5, LD-K**

Ans. Any attempt about the Syntax Errors or Logical Errors. **12 marks**